Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Design of Experiment (DOE) Masterclass

Course Code: C950

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 October 2026	Initial release of the Design of Experiment (DOE) Masterclass lesson plan: two 9:30am – 5:30pm days (15 instructional hours), four topics, ten NovaDOE labs and the v1.0 145-slide deck.	Dr Alfred Ang

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Course Overview

This 2-day, 15-hour course develops practical capability to plan, generate, analyse and evaluate designed experiments using NovaDOE and Excel. Each day runs 9:30am – 5:30pm with 7.5 instructional hours; click-by-click procedures are kept in the Learner Guide.

Learning Outcomes

- LO1: Analyse process interactions to characterise key factors influencing performance in Design of Experiments.
- LO2: Select suitable factorial DoE projects by evaluating relevant performance metrics.
- LO3: Define the scope and execute fractional factorial DoE projects using problem-solving techniques.
- LO4: Evaluate the effectiveness of DoE projects and recommend follow-up actions.

Training Methods and Resources

- Concept-led visual presentation and worked engineering examples.
- NovaDOE browser tool for full factorial, fractional factorial, Plackett-Burman, CCD and space-filling designs.
- Ten individual activity folders with CSV mock data and Excel workbooks.
- Facilitated interpretation, peer review, evidence checks and confirmation planning.
- Learner Guide for all detailed procedures; no click-by-click procedure in the slide deck.

Day 1: Foundations, ANOVA and factorial designs (9:30am – 5:30pm)

Time	Duration	Topic / Activity	Method / Evidence	Slide Mapping
09:30–10:55	85 min	Topic 1: DoE foundations, project framing and measurement	Lecture, visual frameworks and Lab 1	Slides 10–40; Lab 1: slides 36–37
10:55–12:45	110 min	Hypothesis testing, two-sample comparisons and one-way ANOVA	Worked examples and Lab 2	Lab 2: slides 38–39
12:45–13:15	30 min	Lunch	Break	—
13:15–15:05	110 min	Topic 2: 2^2 and 2^3 factorial designs	Effects, interaction, regression; Labs 3–4	Slides 41–75; Lab 3: slides 69–70
15:05–16:30	85 min	ANOVA, centre points,	Worked analysis and	Lab 5: slides 73–74

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		replication and blocking	Lab 5	
16:30–17:30	60 min	Day 1 integration and evidence review	Recap, interpretation and feedback	Admin / recap slides

Day 1 instructional total: 450 minutes (7.5 hours, 9:30am – 5:30pm; the 30-minute lunch is excluded).

Day 2: Fractional screening and optimisation (9:30am – 5:30pm)

Time	Duration	Topic / Activity	Method / Evidence	Slide Mapping
09:30–11:25	115 min	Topic 3: Fractional factorial designs	Generators, aliases, resolution and Lab 6	Slides 77–107; Lab 6: slides 103–104
11:25–12:45	80 min	Screening and sequential experimentation	Plackett-Burman, project selection and Lab 7	Lab 7: slides 105–106
12:45–13:15	30 min	Lunch	Break	—
13:15–14:25	70 min	Screening and sequential experimentation (continued)	Sequential strategy, design review and Lab 7 debrief	Topic 3 recap
14:25–16:15	110 min	Topic 4: RSM, CCD, mixtures and Taguchi	Labs 8–10 and model interpretation	Slides 108–142; Lab 8: slides 136–137
16:15–17:30	75 min	Evaluate effectiveness and recommend follow-up	Confirmation, limitations, recap and course feedback	Wrap-up / feedback slides

Day 2 instructional total: 450 minutes (7.5 hours, 9:30am – 5:30pm; the 30-minute lunch is excluded).

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